DATA ANALYTICS PROJECT PROPOSAL

E-Commerce Sales & Customer Behavior Analysis

An end-to-end data project using SQL · Python · Power BI · Tableau · GitHub

Domain	Difficulty	Duration	Deliverable
E-Commerce / Retail	**Intermediate – Advanced**	**4 – 6 Weeks**	**Dashboard + Report**

1. Project Overview

This project delivers a complete end-to-end data analytics case study using a real-world e-commerce dataset. The work spans the full analytics lifecycle: from raw data ingestion and SQL querying through Python-based statistical analysis and machine-learning-ready feature engineering, culminating in polished, interactive dashboards in Power BI and Tableau. Every artefact is version-controlled on GitHub, making the project portfolio-ready and fully reproducible.

Business Questions Addressed

- Which product categories and regions drive the most revenue?
- How do customer segments behave differently over time?
- Do promotional discounts significantly lift average order value?
- What is the monthly customer retention / churn rate?
- Which customers are at risk of churning (RFM analysis)?

Tools Overview

SQL	Python	Pandas	NumPy	Matplotlib	Statistics	Power BI	Tableau	GitHub

2. Functional Requirements

The table below specifies the ten functional requirements of the project, each mapped to the tool(s) responsible for satisfying it.

ID	Requirement	Description	Tool(s)
FR-01	Data Ingestion	System must load raw CSV/Excel data into a relational SQL database with no data loss.	SQL
FR-02	Data Cleaning	Pipeline must detect and handle nulls, duplicates, and outliers automatically.	Python · Pandas
FR-03	Exploratory Analysis	Notebook must generate at least 10 statistical charts covering distributions & trends.	Python · Matplotlib
FR-04	Statistical Testing	At minimum 3 hypothesis tests with p-values, effect sizes, and conclusions.	Python · Statistics
FR-05	Feature Engineering	Produce RFM table, customer segment labels, and time-based features.	Python · Pandas
FR-06	Power BI Dashboard	Interactive report with 5+ visuals, dynamic slicers, and KPI cards.	Power BI
FR-07	Tableau Story	Published Tableau story with cohort, funnel, and geo views.	Tableau
FR-08	Version Control	All code, queries, and notebooks committed with meaningful messages.	GitHub
FR-09	Documentation	README with setup instructions, data dictionary, and findings summary.	GitHub · Jupyter
FR-10	Reproducibility	Any reviewer can clone the repo and reproduce all results end-to-end.	GitHub · Python

3. Step-by-Step Project Guide

Each phase below represents a concrete workstream with clear objectives, key activities, and the tool(s) used. Follow phases sequentially; each builds on the output of the previous one.

#	Phase	Key Activities	Tools
1	Business Understanding & Setup	Define KPIs, repo structure, data discovery	GitHub
2	Data Acquisition & Storage	Download datasets, load to SQL database, write queries	SQL · GitHub
3	Data Cleaning & EDA	Handle nulls, outliers, describe distributions, plot histograms	Python · Pandas · NumPy · Matplotlib
4	Descriptive Statistics	Mean, median, std dev, skewness, correlation matrices	Python · Statistics
5	Probability & Inference	Confidence intervals, A/B test on campaigns, chi-square tests	Python · Statistics
6	Feature Engineering	RFM scoring, date features, customer segments, encoding	Python · Pandas · Feature Eng.

7	Hypothesis Testing	Test whether discounts significantly affect revenue, ANOVA	Python · Statistics
8	Interactive Dashboard (Power BI)	KPI cards, time-series sales, map by region, slicers	Power BI
9	Advanced Visualization (Tableau)	Customer journey funnel, cohort heatmap, story points	Tableau
10	Documentation & Version Control	README, Jupyter notebooks, push all artefacts	GitHub

Phase Details

Phase 1 – Business Understanding & GitHub Setup

- Create a GitHub repository with folders: /data/raw, /data/clean, /sql, /notebooks, /dashboards. ***[GitHub]***
- Write a README with project goals, KPI definitions, and setup instructions. ***[GitHub]***
- Set up .gitignore to exclude large raw files; use Git LFS if needed. ***[GitHub]***

Phase 2 – Data Acquisition & SQL Layer

- Download the chosen dataset(s) from Kaggle / UCI (see Section 5). ***[SQL]***
- Create a local PostgreSQL or SQLite database and load CSVs using COPY or pandas.to_sql(). ***[SQL · Python]***
- Write SQL queries for monthly sales totals, top 10 products, customer order counts. ***[SQL]***
- Export aggregated query results as CSVs for downstream Python consumption. ***[SQL]***

Phase 3 – Data Cleaning & EDA

- Load raw data with pd.read_csv(); inspect dtypes, null counts, and duplicates. ***[Python · Pandas]***
- Impute or drop nulls; cap outliers using IQR method; parse date columns. ***[Python · Pandas · NumPy]***
- Plot histograms, boxplots, and time-series line charts for all key numeric variables. ***[Python · Matplotlib]***
- Compute a Pearson correlation matrix and visualize as a heatmap. ***[Python · Matplotlib]***

Phase 4 – Descriptive Statistics

- Report mean, median, mode, std dev, skewness, and kurtosis for revenue, quantity, and price. ***[Python · Statistics]***
- Aggregate monthly KPIs: total revenue, average order value, unique customers. ***[Python · Pandas]***
- Build a probability distribution plot to understand order value spread. ***[Python · Statistics · Matplotlib]***

Phase 5 – Probability & Inferential Statistics

- Calculate 95% confidence intervals for mean order value per customer segment. ***[Python · Statistics]***
- Conduct a two-proportion z-test comparing conversion rates before vs. after a promotion. ***[Python · Statistics]***
- Report p-values and interpret practical vs. statistical significance. ***[Python · Statistics]***

Phase 6 – Feature Engineering

- Build RFM (Recency, Frequency, Monetary) scores and label customers as Champions, At-Risk, etc. ***[Python · Pandas · Feature Eng.]***
- Extract day-of-week, month, quarter, and is_weekend from order date. ***[Python · Pandas]***
- Apply label encoding to categorical product categories for downstream modelling readiness. ***[Python · Pandas]***

Phase 7 – Hypothesis Testing

- H0: Discounts do not affect average revenue per order. Run independent t-test and report result. ***[Python · Statistics]***
- H0: Customer segment has no effect on return rate. Run one-way ANOVA. ***[Python · Statistics]***
- Interpret effect size (Cohen's d) alongside p-values to avoid over-reliance on significance. ***[Python · Statistics]***

Phase 8 – Power BI Dashboard

- Import cleaned SQL export into Power BI Desktop; build a star schema in the data model. ***[Power BI]***
- Create KPI cards (Total Revenue, Orders, Avg Order Value, Active Customers). ***[Power BI]***
- Add time-series area chart, bar chart by category, map by region, and RFM segment donut. ***[Power BI]***
- Publish to Power BI Service and share a public link. ***[Power BI]***

Phase 9 – Tableau Story

- Connect Tableau Desktop to the cleaned CSV or SQL database. ***[Tableau]***
- Build a cohort retention heatmap, customer funnel, and revenue geo map. ***[Tableau]***
- Assemble a 4-slide Tableau Story with narrative captions for each insight. ***[Tableau]***
- Publish to Tableau Public and embed the link in the GitHub README. ***[Tableau · GitHub]***

Phase 10 – Documentation & Final Commit

- Finalize all Jupyter notebooks with markdown cells explaining each step. ***[Jupyter · GitHub]***
- Tag the repo release (v1.0) and write a release summary. ***[GitHub]***
- Export a PDF report of key findings from Jupyter using nbconvert. ***[Python · Jupyter]***

4. Presentable Results

Upon completion, the following outputs will be ready for portfolio presentation or stakeholder delivery:

4.1 Deliverables

Deliverable	Contents	Tools
SQL Scripts	DDL + DML queries for ingestion, cleaning, aggregation	SQL · GitHub
Jupyter Notebooks	EDA, statistics, hypothesis testing, feature engineering notebooks	Python · Jupyter
Power BI .pbix File	Interactive sales & customer KPI dashboard	Power BI

Tableau Workbook	Customer journey story & cohort analysis	Tableau
GitHub Repository	Full project with README, data dictionary, version history	GitHub
Final PDF Report	Executive summary with insights, charts, and recommendations	Python · Matplotlib

4.2 Key Insights Expected

- Identification of the top 3 revenue-generating product categories with seasonal trends.
- Customer segmentation revealing the proportion of Champions vs. At-Risk customers.
- Statistical confirmation (or rejection) of whether discount campaigns are effective.
- Monthly retention curve showing cohort drop-off patterns.
- Geo-distribution of orders highlighting high-opportunity untapped regions.

4.3 Presentation Format

- Power BI dashboard shared via a public Power BI Service link.
- Tableau Public story accessible via URL (embeddable in a portfolio website).
- GitHub repository linked on your professional profile (LinkedIn / resume).
- PDF executive summary (2–3 pages) for non-technical stakeholders.

5. Data Sources

All datasets are freely available and do not require institutional access. Download, document provenance in the README, and commit a sample (first 1,000 rows) to the repository for reproducibility.

Source	Description	URL	Role
Kaggle – E-Commerce Data	Actual transactions of a UK retailer (541K rows)	https://www.kaggle.com/datasets/carrie1/ecommerce-data	Primary
Kaggle – Brazilian E-Commerce (Olist)	100K orders with products, payments, reviews	https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce	Primary
UCI ML Repository – Online Retail II	UK online retail 2009-2011	https://archive.ics.uci.edu/dataset/502/online+retail+ii	Primary

Statista / World Bank	Macro e-commerce benchmarks	https://www.statista.com / https://data.worldbank.org	Supplemental

Recommended Primary Dataset

Start with the Kaggle E-Commerce Data (UK retailer) as it is clean enough to begin quickly yet rich enough to support all planned analyses. Supplement with the Olist dataset for geo-diversity and review sentiment if time allows.

6. Suggested Timeline

- Week 1 – Phases 1 & 2: Setup, SQL ingestion, initial queries.
- Week 2 – Phases 3 & 4: Cleaning, EDA, descriptive statistics.
- Week 3 – Phases 5 & 6: Inferential stats, feature engineering.
- Week 4 – Phase 7: Hypothesis testing and write-up.
- Week 5 – Phases 8 & 9: Power BI and Tableau dashboards.
- Week 6 – Phase 10: Documentation, GitHub polish, final review.

7. Success Criteria

The project is considered complete and portfolio-ready when all of the following conditions are met:

- All 10 functional requirements (Section 2) are demonstrably satisfied.
- Power BI and Tableau links are publicly accessible with no data errors.
- GitHub repository has at least 15 meaningful commits spanning all phases.
- At least 3 actionable business recommendations are supported by statistical evidence.
- A peer reviewer can clone the repo and reproduce all results in under 30 minutes.

Ready to build something impressive?

Fork the repo, follow the phases, and let the data tell the story.